Pythia-160M Seed-9 Alignment Transfer

Repeat-match causal head roles across all official seeds

Branch specialization research project

2026-05-22

Checkpoint reason: this is the full 9-seed replication of the Pythia-160M repeat-match alignment-transfer result.

Question

The 3-seed Pythia-160M run suggested that raw-score alignment transfers repeat-match causal head roles much better than same head index.

This checkpoint asks:

does that aligned-transfer result survive all 9 official Pythia-160M seeds?

This is the current strongest test of weak, relabeled cross-seed role universality in the project.

Setup

- Models: `EleutherAI/pythia-160m-seed1` through seed9
- Checkpoints: step4000, step16000, step143000
- Layers: 0 and 1
- Selected heads: top repeat-match head per tested layer
- Alignment: raw attention-score Hungarian matching using all 8 fixed probe texts
- Causal test: zero selected head outputs; measure repeated-token continuation loss
- Transfer pairs: 72 ordered source-target seed pairs per checkpoint

Metric Glossary

Probe specialization Normalized repeat-match attention pattern score.

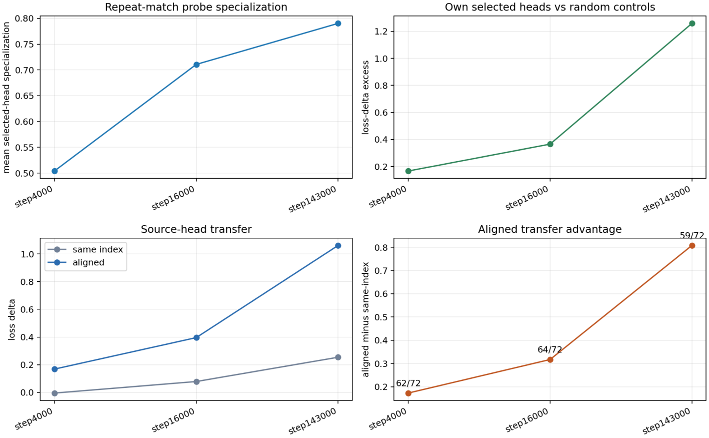
Own top minus random Target seed selected-head loss delta minus random same-layer controls.

Same-index transfer Ablate target seed heads at the source seed's raw layer/head indices.

Aligned transfer Ablate target seed heads matched to source heads by raw-score alignment.

Aligned better Number of source-target pairs where aligned transfer has larger loss delta than same-index transfer.

Trajectory



Alignment transfer is already detectable by step4000 and grows strongly by the final checkpoint.

Aggregate Results

Revision	Probe	Own − rand.	Same-index	Aligned	Align − same	Better
step4000	0.5039	0.1659	-0.0049	0.1682	0.1731	62/72
step16000	0.7107	0.3650	0.0785	0.3958	0.3173	64/72
step143000	0.7903	1.2586	0.2541	1.0619	0.8078	59/72

Means over 9 seeds. Transfer means use 72 ordered source-target seed pairs per checkpoint.

What Changed From The 3-Seed Pilot

Confirmed

- Raw-score alignment transfers causal repeat-match roles much better than raw same-index transfer.
- Same-index transfer remains weak at every selected checkpoint.

Revised

- The 3-seed pilot made step4000 look mostly probe-only.
- The 9-seed run shows measurable aligned causal transfer already at step4000: aligned-minus-same is 0.1731 and aligned wins in 62/72 pairs.

Interpretation

Supported

- Repeat-match role identity is weakly universal across Pythia-160M seeds after relabeling by raw-score alignment.
- Causal role transfer strengthens through training: aligned-minus-same rises from 0.1731 to 0.8078.
- Probe specialization, causal importance, and aligned transfer all strengthen across training, but not in a clean all-or-nothing sequence.

Not shown

- This does not prove branch-level functional modularity in ordinary Pythia attention heads.

Caveats

- Only Pythia-160M is covered in the 9-seed scale-up.
- Only layers 0 and 1 were tested.
- Synthetic repeated-token continuation task.
- Head-output zero ablation, not full path patching.
- Alignment uses 8 short probe texts; a larger probe corpus remains a useful robustness check.
- The first all-checkpoint run was interrupted by unrelated GPU contention, so the final result uses three completed one-checkpoint jobs.

Decision And Next Step

Decision. Treat aligned causal transfer as a validated Phase 1 metric for repeat-match heads in Pythia-160M.

Next step. Test whether this is specific to repeat-match heads:

- add a second causal head-role task, such as previous-token or induction-style copying;
- keep same-index transfer as the negative baseline;
- add path patching after a second role survives head-output ablation.

Provenance

Source

- `scripts/pythia_repeat_match_alignment_trajectory.py`
- `scripts/analyze_pythia_alignment_trajectory.py`
- `scripts/analyze_pythia_seed9_alignment_selected.py`
- `doc/phase1_pythia160m_seed9_alignment_selected_checkpoints.md`

Copied data

- `../data/revision_summary.csv`
- `../data/condition_summary.csv`
- `../data/transfer_pair_summary.csv`
- `../figures/seed9_alignment_selected_trajectory.png`